

A kernel counterexample and the canonical repair for the reverse Laplace polynomial inequality

Candidate counterexample and strengthened full result for arXiv:2502.07448

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Status. Candidate full counterexample, likely valid, requiring human review. The printed conjecture is false because its left side annihilates constants while its right side does not. More substantially, the canonical repair obtained by taking the right side modulo additive constants is true, with both inequality directions.

1 The printed conjecture and its repair

Let

$$d\mu(x) = \frac{1}{2}e^{-|x|} dx$$

and let $(P_k)_{k \geq 0}$ be its orthonormal polynomials, with $P_0 = 1$. For $f \in L^2(\mu)$ put

$$\mathcal{S}_\mu(f) = \sum_{k=1}^{\infty} \log^2(e+k) |\langle f, P_k \rangle_{L^2(\mu)}|^2. \quad (1)$$

Bizeul–Klartag prove an upper bound on (1) and then conjecture the reverse inequality shown below [1, p. 3].

but

$$\sum_{k=1}^{\infty} a_k \log^2(e+k) (f_a)_k^2 = +\infty$$

We conjecture that inequality (3) may be reversed, and in fact for any absolutely continuous function f ,

$$\sum_{k=1}^{\infty} \log^2(e+k) f_k^2 \gtrsim \int_{\mathbb{R}} \log^2(e+|x|) f^2 d\mu_1 + \int_{\mathbb{R}} (f')^2 d\mu_1.$$

Comparing to (2), we get a stronger control on the coefficients f_k , showing that the sequence

Figure 1: The conjecture in the source PDF. Its quantifier is “any absolutely continuous function f ”.

Thus the literal assertion is that, for some universal $c > 0$ and every absolutely continuous f ,

$$\mathcal{S}_\mu(f) \geq c \left(\int_{\mathbb{R}} \log^2(e+|x|) |f(x)|^2 d\mu(x) + \int_{\mathbb{R}} |f'(x)|^2 d\mu(x) \right). \quad (2)$$

Proposition 1 (Counterexample to the printed conjecture). *The inequality (2) is false.*

Proof. Take $f \equiv 1$. Orthogonality gives $\langle 1, P_k \rangle = 0$ for every $k \geq 1$, and $f' = 0$, whereas

$$\int_{\mathbb{R}} \log^2(e + |x|) d\mu(x) > 0.$$

Hence the two sides of (2) are respectively zero and strictly positive. $\square$

The natural invariant replacement of the right side is

$$\mathcal{Q}_\mu(f) = \inf_{a \in \mathbb{K}} \int_{\mathbb{R}} \log^2(e + |x|) |f(x) - a|^2 d\mu(x) + \int_{\mathbb{R}} |f'(x)|^2 d\mu(x). \quad (3)$$

Here and below $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$. This is not merely a way to remove the counterexample: it gives the exact corrected theorem.

Theorem 2 (Two-sided canonical repair). *There are universal constants $0 < c \leq C < \infty$ such that every absolutely continuous f satisfies*

$$c \mathcal{Q}_\mu(f) \leq \mathcal{S}_\mu(f) \leq C \mathcal{Q}_\mu(f), \quad (4)$$

with the usual extended-value interpretation. Equivalently, the printed conjecture becomes true after quotienting both sides by constants.

Proof intuition

The source replaces the Laplace density, up to a dilation and bounded density perturbation, by the hyperbolic-secant density. There the orthonormal Meixner–Pollaczek polynomials have an explicit generating function. Its x -derivative turns differentiation into a one-sided Hilbert matrix; a dyadic near/far split costs exactly one logarithm. Multiplication by x becomes the Jacobi matrix with off-diagonal entries $n + 1$. A short limiting interpolation argument then transfers the graph bound for this Jacobi matrix to a logarithmic graph bound. These two estimates prove the new direction $\mathcal{Q} \lesssim \mathcal{S}$; the other direction is the source theorem applied after subtracting an arbitrary constant.

2 The hyperbolic-secant model

Let

$$d\nu(x) = \frac{dx}{2 \cosh(\pi x/2)}. \quad (5)$$

The source records the generating function

$$G(x, z) = \frac{e^{x \arctan z}}{\sqrt{1 + z^2}} = \sum_{n=0}^{\infty} p_n(x) z^n, \quad |z| < 1, \quad (6)$$

where $(p_n)_{n \geq 0}$ is an orthonormal basis of $L^2(\nu)$ [1, Sec. 1.1]. Differentiating (6) in z and in x gives

$$x p_n = (n + 1) p_{n+1} + n p_{n-1}, \quad (7)$$

$$p'_n = \sum_{\substack{1 \leq r \leq n \\ r \text{ odd}}} \frac{(-1)^{(r-1)/2}}{r} p_{n-r}. \quad (8)$$

We use these identities to control both terms in (3).

Lemma 3 (Derivative estimate). *For every finite scalar sequence (c_n) ,*

$$\left\| \left(\sum_{n \geq 0} c_n p_n \right)' \right\|_{L^2(\nu)}^2 \leq C \sum_{n \geq 1} \log^2(e+n) |c_n|^2. \quad (9)$$

Proof. By (8) and orthonormality, the coefficient of p_m in the derivative is

$$(Tc)_m = \sum_{\substack{n > m \\ n-m \text{ odd}}} \frac{\varepsilon_{n-m}}{n-m} c_n, \quad |\varepsilon_{n-m}| = 1.$$

Let $c^{(j)}$ be the restriction of c to $I_j = [2^j, 2^{j+1}) \cap \mathbb{N}$, for $j \geq 0$. Split $Tc^{(j)}$ into the near part $m \geq 2^{j-1}$ and the far part $m < 2^{j-1}$, with the evident convention for $j = 0$.

For the near part, Young's convolution inequality and the harmonic-sum bound give

$$\|(Tc^{(j)})_{\text{near}}\|_2 \leq \left(\sum_{1 \leq r \leq 2^{j+1}} \frac{1}{r} \right) \|c^{(j)}\|_2 \leq C(j+1) \|c^{(j)}\|_2.$$

Its output is supported in $[2^{j-1}, 2^{j+1})$, so these output supports have uniformly bounded overlap as j varies. Hence the squared norm of the sum of all near parts is at most $C \sum_j (j+1)^2 \|c^{(j)}\|_2^2$.

For the far part, its Hilbert–Schmidt norm is uniformly bounded because

$$\sum_{n \in I_j} \sum_{m < 2^{j-1}} \frac{1}{(n-m)^2} \leq C.$$

Therefore

$$\begin{aligned} \left\| \sum_j (Tc^{(j)})_{\text{far}} \right\|_2 &\leq C \sum_j \|c^{(j)}\|_2 \\ &\leq C \left(\sum_j (j+1)^2 \|c^{(j)}\|_2^2 \right)^{1/2}, \end{aligned}$$

where the last step uses $\sum_j (j+1)^{-2} < \infty$. Since $j+1 \asymp \log(e+n)$ on I_j , combining the two parts proves (9). $\square$

We next isolate the logarithmic interpolation fact used for the spatial weight.

Lemma 4 (Logarithmic graph interpolation). *Let $A, B \geq I$ be positive self-adjoint operators on a Hilbert space. If $D(B) \subseteq D(A)$ and*

$$\|Au\| \leq M \|Bu\| \quad (u \in D(B)), \quad (10)$$

then

$$\|(I + \log A)u\| \leq C_M \|(I + \log B)u\|. \quad (11)$$

Proof. For $T \geq I$, use the quadratic K -functional

$$K_T(t, u) = \inf_{u=u_0+u_1} (\|u_0\|^2 + t^2 \|Tu_1\|^2)^{1/2}.$$

The spectral theorem and scalar minimization give

$$K_T(t, u)^2 = \int_{[1, \infty)} \frac{t^2 \lambda^2}{1 + t^2 \lambda^2} d\|E_T(\lambda)u\|^2.$$

For every $\lambda \geq 1$, splitting the t -integral at $1/\lambda$ shows

$$1 + \int_0^1 \log(1/t) \frac{t^2 \lambda^2}{1 + t^2 \lambda^2} \frac{dt}{t} \asymp (1 + \log \lambda)^2. \quad (12)$$

Thus the integral of $K_T(t, u)^2$ in (12), plus $\|u\|^2$, is equivalent to $\|(I + \log T)u\|^2$.

Condition (10) gives $K_A(t, u) \leq K_B(Mt, u)$. Substitute $s = Mt$ in the integral. On $0 < s < 1$, the extra constant $\log M$ is controlled by the same integral plus $\|u\|^2$; on $1 \leq s \leq M$, use $K_B(s, u) \leq \|u\|$. Applying (12) for A and B proves (11). $\square$

Lemma 5 (Spatial logarithm). *For every finite scalar sequence (c_n) ,*

$$\int_{\mathbb{R}} \log^2(e + |x|) \left| \sum_{n \geq 0} c_n p_n(x) \right|^2 d\nu(x) \leq C \sum_{n \geq 0} \log^2(e + n) |c_n|^2. \quad (13)$$

Proof. On $\ell_2(\mathbb{N}_0)$ let J be the Jacobi operator corresponding to (7), and let $Be_n = (n + 1)e_n$. Thus the polynomial transform takes J to multiplication by x . Directly from the two shifts,

$$\|Jc\|_2^2 \leq 4\|Bc\|_2^2.$$

Set $A = (I + J^2)^{1/2}$. Then $A, B \geq I$ and

$$\|Ac\|_2^2 = \|c\|_2^2 + \|Jc\|_2^2 \leq 5\|Bc\|_2^2.$$

Lemma 4 applies. Finally, on the spectral side, the scalar functions

$$\log(e + |x|), \quad 1 + \log \sqrt{1 + x^2}$$

are comparable, while on coefficients $1 + \log(n + 1) \asymp \log(e + n)$. This is exactly (13). $\square$

3 Return to the Laplace measure

We use an elementary comparison principle already exploited in the source. If probability measures α and β satisfy $c\beta \leq \alpha \leq C\beta$, then their best polynomial approximation errors are comparable. If (a_k^α) and (a_k^β) are the coefficients of the same f in their respective orthonormal polynomial bases, then

$$\sum_{k \geq n} |a_k^\alpha|^2 \asymp \sum_{k \geq n} |a_k^\beta|^2.$$

For any increasing sequence w_k with $w_0 = 0$, summation by parts gives

$$\sum_{k \geq 1} w_k |a_k|^2 = \sum_{n \geq 1} (w_n - w_{n-1}) \sum_{k \geq n} |a_k|^2.$$

Consequently the coefficient sums with $w_k = \log^2(e + k)$ are comparable.

Under the dilation $y = \pi x/2$, the measure (5) becomes

$$d\rho(y) = \frac{dy}{\pi \cosh y}.$$

Its density and $\frac{1}{2}e^{-|y|}$ are bounded above and below by universal multiples of each other. The dilation and this density comparison also preserve, up to universal constants, the derivative norm and

$$\inf_a \int_{\mathbb{R}} \log^2(e + |y|) |f(y) - a|^2 d\rho(y).$$

Proof of Theorem 2. In the ν model, subtract the constant coefficient c_0 from $f = \sum c_n p_n$. Lemmas 3 and 5 give

$$\inf_a \int \log^2(e + |x|) |f - a|^2 d\nu + \int |f'|^2 d\nu \leq C \sum_{n \geq 1} \log^2(e + n) |c_n|^2.$$

The dilation and comparison principle transfer this inequality to μ , proving the left inequality in (4). The estimates extend from polynomials by completion; multiplication by the spatial weight and weak differentiation are closed operators.

For the other direction, apply Theorem 1 of the source to $f - a$. Its positive-degree coefficients and derivative do not depend on a , so taking the infimum over a proves $\mathcal{S}_\mu(f) \leq C \mathcal{Q}_\mu(f)$. $\square$

4 Verification, sharpness, and novelty boundary

The source’s unpublished reverse-proof draft, still present as commented source code in arXiv:2502.07448v1, contains the false step

$$\sum_{k \geq 1} |f_k|^2 \geq \int |f|^2 d\nu,$$

which is precisely the missing constant coefficient exposed by Proposition 1. The June 17, 2025 authors’ version and the December 2025 journal publication still print the literal conjecture on PDF page 3.

As a non-proof stress test, the Jacobi and derivative matrices were truncated at five doubling dimensions from 16 through 256, with 512 padding modes for spectral quadrature. The largest generalized eigenvalue of the repaired right side against the logarithmic coefficient form stayed between 1.9954 and 1.9956. At dimension 512 the spatial and derivative pieces separately had largest ratios about 1.6494 and 0.7704. The reusable script is included as `code/finite_section_probe.py`.

There is also an exact high-frequency check. For $f_t(x) = e^{itx}$ under ν , (6) and the transform $\int e^{ux} d\nu(x) = 1/\cos u$ give

$$\langle f_t, p_n \rangle = \operatorname{sech}(t) [i \tanh(t)]^n.$$

Thus the squared coefficients form a geometric law of typical degree $e^{2|t|}$, and their logarithmic energy is comparable to t^2 , matching $\|f'_t\|_2^2 = t^2$. This confirms the sharp logarithmic scale.

The four run indexes had no prior entry for this paper or conjecture. Bounded searches through 11 August 2026 used the arXiv id, the exact conjecture phrase, the title with “reverse inequality”, the authors with “Meixner–Pollaczek”, and later citations. They found no correction or answer. Both the constant observation and the repaired theorem could be folklore or independently known; no priority claim is made.

Human-review recommendation. First confirm that the published quantifier really includes constants. For the strengthened theorem, independently check the K -functional identity (12), the scale change in Lemma 4, and the coefficient-tail comparison under bounded density perturbations. The dyadic derivative estimate is otherwise elementary.

References

- [1] P. Bizeul and B. Klartag, *Polynomial approximation in L^2 with the double-sided exponential weight via complex analysis*, Journal of Approximation Theory **312** (2025), 106218; arXiv:2502.07448v1.